

Spring Boot Learning Roadmap

Structured Index for Learning and Interview Preparation

Phase 1 - Spring Foundations

- Java & OOP Refresher
- IoC and Dependency Injection
- Beans and Bean Scopes
- Spring Core Annotations

Phase 2 - Spring Boot Essentials

- Spring Initializr
- Auto Configuration
- Starter Dependencies
- Project Structure

Phase 3 - Configuration Management

- Properties and YAML
- Profiles
- Externalized Configuration
- Configuration Properties

Phase 4 - Web Development

- MVC Architecture
- Controllers
- Request/Response Handling
- Validation
- Global Exception Handling

Phase 5 - Database Layer

- JPA
- Hibernate
- Entity Relationships
- Transactions
- Pagination and Sorting

Phase 6 - RESTful Services

- CRUD APIs
- DTOs
- ResponseEntity
- API Documentation Swagger/OpenAPI

Phase 7 - Security

- Spring Security Basics
- JWT
- OAuth2
- Role Based Access

Phase 8 - Testing

- JUnit 5
- Mockito
- Integration Testing
- Testcontainers

Phase 9 - Microservices

- Eureka
- Config Server
- Feign Client
- API Gateway
- Resilience4j

Phase 10 - Event Driven Systems

- Kafka
- RabbitMQ
- Messaging Patterns

Phase 11 - Production Readiness

- Actuator
- Logging
- Metrics
- Monitoring

Phase 12 - Deployment

- Docker
- Kubernetes
- CI/CD
- Cloud Platforms

Phase 13 - Advanced Topics

- AOP
- Caching
- Scheduling

- Async
- WebFlux

Phase 14 - Interview Track

- Frequently Asked Questions
- Architecture Scenarios
- Best Practices